

Sign Language to Speech: The Role of Conversion Gloves in Enhancing Communication

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Abstract—Individuals with speech disabilities often rely on sign language for communication; however, a significant challenge arises when interacting with people who are unfamiliar with it, creating a communication barrier. This study proposes a cost-effective solution to address this issue by developing smart gloves that enable speech-impaired individuals to communicate more effectively. The system utilizes gloves embedded with flex and motion sensors to detect various sign language gestures. These sensor readings are processed using an Arduino UNO, which transmits the data to an Android device via a Bluetooth module. Gesture recognition is based on pre-defined patterns corresponding to specific sign language gestures, ensuring efficient processing with minimal latency. While the current prototype focuses on American sign language (ASL), it can be adapted for other sign language variants with appropriate calibration. The system aims to bridge the communication gap by translating users' gestures into text and converting the text into speech, providing a practical and portable solution for real-world scenarios. Challenges related to gesture recognition accuracy, sign language variations, and real-time translation latency are acknowledged, with potential improvements discussed for future work.

Index Terms—Sign language, ISL, ASL, gesture

I. INTRODUCTION

Communication is a necessary aspect of human interaction, yet for individuals with hearing impairments, conveying ideas and emotions through traditional speech becomes a challenge [1]. Sign language, a visual form of communication, empowers individuals with hearing impairments to express themselves easily. However, the limited availability of sign language interpreters and the challenges of cross-modal communication have often hindered the full integration of the deaf and mute communities into society. Bridging this gap is a technological innovation that promises to revolutionize sign language interpretation: conversion gloves.

Sign language conversion gloves are specialized wearable devices that aim to convert sign language gestures into audible speech [2], [3]. This technology holds promise for connect-

ing communication barriers between the deaf and hearing communities, promoting inclusivity and mutual perception. Equipped with sensors, these gloves capture hand gestures and movements, enabling real-time translation into spoken words. As highlighted in [4], research has demonstrated the ability of sign language recognition systems to classify Bangla digits, signs, and alphabets efficiently. However, challenges remain in generating coherent sentences and speech output.

Researchers have examined the amalgamation of information technology and communication tools, including smartphones and robotic hands, to develop more advanced translation systems [5], [6]. The need to address persistent communication barriers experienced by those who rely on sign language as their major form of expression is what motivates this endeavour. This study intends to promote additional innovation and acceptance of technology that bridge communication gaps and enhance the quality of life for individuals who gain from such breakthroughs by investigating the design, functioning, and potential of sign language conversion gloves.

The objective of this work is to design and integrate advanced technology into a glove equipped with sensors such as flex sensors, accelerometers, and touch sensors to accurately detect and track finger movements, hand gestures, and posture. By leveraging embedded technologies and IoT devices, the system aims to enable real-time gesture recognition for the accurate translation of sign language into text or speech [7]. The salient contributions of the work include:

- Introduces a cost-effective, portable system for sign language conversion using sensor-equipped gloves to translate gestures into text and speech.
- Leverages an Arduino UNO, Bluetooth module, and Android application to enable real-time gesture recognition and output.
- Integrates flex sensors and IMUs to detect hand movements with high precision, improving gesture accuracy.

Section II discusses the related work, while Section III highlights the mathematical modeling of the flex sensor. Section IV presents the proposed system along with experimental results, and conclusions are provided in Section V.

II. RELATED WORK

Researchers have employed computer vision algorithms to analyze video streams of sign language gestures. The study in [8] proposed a glove-based approach to recognize the Indian sign language (ISL) that was designed to improve communication for hearing and speech impaired individuals. The significance of the sign language as communication medium was highlighted. The proposed system utilized an economically feasible webcam and voice board to capture and analyse ISL gestures, which is then converted into text and speech formats. The system focused on minimizing the communication gap between those unfamiliar of sign language and hearing-impaired individuals.

The research in [9] presented an automated hand gesture robot designed to enhance communication between those with hearing problems and those with normal hearing. Flex sensors installed on a glove were used by the system to record hand movements, and the data was wirelessly sent via Bluetooth to a robotic hand. About 30 meters was its operational range. Similarly, [10] suggested a system that uses artificial neural networks and a sensory glove to recognise the ASL. Flex sensors were included into the hardware to measure the angles of the finger joints, and trained neural networks were used in MATLAB to identify ASL letters. During the ANN training process, several readings were taken for every letter in order to improve accuracy.

The development of a wearable smart glove equipped with inertial measurement units (IMUs) to recognize hand gestures was documented in [11]. The design was aimed to improve human-computer interaction by providing a chance for users to interact with digital surroundings in three dimensions and applications of the ASL. The real time readings were tracked by the glove, which then was processed using various machine learning techniques. The outcome showed precision in recognizing the movements. In [12], a hand gesture recognition system was built to convert ISL into speech and text. The system used a sensor-based approach, which employed the flex sensors mounted on the gloves to track hand gestures, which then was processed by an Arduino UNO microcontroller. The gestures that were recognized were sent to an android application, which enabled to display real time data and vocalization of the translated signs.

The work in [13] capture the hand gestures using webcam and the output was converted in the form of text as well as voice, making communication much easier for deaf and mute individuals. It processed the captured images through several stages viz., preprocessing, feature extraction, and classification. The convolutional neural networks were used to extract features from hand gestures and classify them into corresponding signs. A smart glove to bridge the communication gap between hearing-impaired people and the general population

was proposed in [14]. The system utilized Arduino UNO microcontroller with flex sensors and accelerometers integrated into gloves to detect and interpreted the ASL gestures output into audible speech through earphones. The flex sensors were used to measure finger movements and convert the same into resistance values, which was further processed using Arduino to convert the analog signals into digital output. The system compared this digital output with pre-loaded ASL dataset to recognize and interpret both individual alphabets and complete words.

The article in [15] discussed the progression of a wearable glove designed to translate the ASL into text and speech, aimed to support the communication for deaf individuals. The system utilized flex sensors to detect the finger movements, which were processed by an Arduino nano microcontroller and converted the gestures into digital signals. These signals were then converted to speech through text-to-speech technology. An approach to sign language translation using transfer learning to address the data unavailability issue inherent in existing sign language datasets was discussed in [16]. The proposed method achieved state-of-the-art performance on two sign language translation benchmarks, surpassing previous methods significantly.

The work in [17] presented a system that interpreted hand motions from the ASL alphabet using a wearable glove. The system used slide potentiometers and force-sensitive resistors to detect finger movements. A neural network was created and trained using Google Collab and Python, achieving accuracy rates of 93%, 95%, and 97% across three different datasets with varying numbers of samples. The system was tested in real-time and successfully displayed the corresponding letters on a computer screen. A smart glove system that combined strain-sensitive conductive knit sensors and an accelerometer to detect and distinguish the ASL gestures in real-time was proposed in [18]. This machine featured sixteen resistive sensors spanning hand moves and used an STM32 microcontroller for on-board processing. The researchers developed a neural network pipeline capable of recognizing 24 alphabetical letters and phrases, which include each static poses and dynamic gestures. Table I shows a comparison of the literature on sign language recognition and translation.

III. MATHEMATICAL MODEL OF FLEX SENSOR

A flex sensor measures the degree of bending or flexion by producing a change in resistance. The resistance of the flex sensor, R_f , is dependent on the bending angle θ :

$$R_f(\theta) = R_0 + k \cdot \theta \quad (1)$$

where $R_f(\theta)$: Resistance of the flex sensor at a given bending angle θ (in ohms), k : Sensitivity constant of the flex sensor (in ohms per degree), R_0 : Resistance of the flex sensor when unbent (in ohms), and θ : Bending angle (in degrees). When used in a voltage divider circuit, the output voltage V_{out} is given by:

$$V_{out} = \frac{R_f(\theta)}{R_f(\theta) + R_s} \cdot V_{in} \quad (2)$$

TABLE I: Comparison of literature on sign language recognition and translation.

Ref.	Sign Language	Hardware Used	Techniques/Algorithms	Key Outcomes
[8]	ISL	Webcam, Voice Board	Image processing to capture gestures	Converted ISL gestures into text and speech, improved communication for hearing impaired.
[10]	ASL	Sensory Glove with Flex Sensors	Artificial Neural Networks (ANN) in MATLAB	Recognized ASL letters with trained ANN using multiple readings for accuracy.
[9]	Hand Gesture Robot	Glove with Flex Sensors, Bluetooth	Data transmission to robotic hand	Enabled gesture-based robotic hand operation within a 30-meter range.
[11]	ASL	Smart Glove with IMUs	Machine Learning Techniques	Real-time 3D gesture recognition, improved human-computer interaction.
[12]	ISL	Flex Sensors, Arduino UNO, Android Application	Sensor-based tracking and real-time processing	Converted ISL gestures to text and speech via Android app.
[13]	General Hand Gestures	Webcam	Convolutional Neural Networks (CNN)	Processed gestures into text and voice, aiding communication for deaf and mute.
[14]	ASL	Glove with Flex Sensors and Accelerometers, Arduino UNO	Dataset comparison with pre-loaded ASL	Translated ASL gestures into audible speech and text output.
[15]	ASL	Flex Sensors, Arduino Nano	Text-to-Speech Technology	Converted gestures into speech for effective communication.
[16]	Sign Language Translation	Transfer Learning	Transfer Learning-based on limited datasets	Achieved state-of-the-art performance on translation benchmarks.
[17]	ASL Alphabet	Slide Potentiometers, Force-Sensitive Resistors	Neural Network trained with Google Collab	Achieved high accuracy (93-97%) for letter recognition and real-time display.
[18]	ASL	Knit Sensors, Accelerometer, STM32 Microcontroller	Neural Network Pipeline	Recognized static and dynamic gestures for 24 letters and words in real-time.

where V_{in} : Input voltage applied to the circuit (in volts), V_{out} : Output voltage of the voltage divider (in volts), and R_s : Series resistor value (in ohms).

To map the bending of the sensor to a gesture, the angle θ is calculated using the measured resistance R_f :

$$\theta = \frac{R_f - R_0}{k} \quad (3)$$

This angle is then used to identify gestures based on predefined thresholds:

$$\text{Gesture} = \begin{cases} \text{Gesture 1,} & \text{if } \theta_1 \leq \theta < \theta_2, \\ \text{Gesture 2,} & \text{if } \theta_2 \leq \theta < \theta_3, \\ \vdots & \\ \text{Gesture N,} & \text{if } \theta_{N-1} \leq \theta \leq \theta_N \end{cases} \quad (4)$$

The angular coordinates-pitch, roll, and yaw-are derived from IMU module data to estimate the hand's position in three-dimensional space. The IMU module provides accelerometer and gyroscope data, which are used to calculate the angular coordinates:

- Pitch (θ_x): The rotation around the x-axis, calculated as:

$$\theta_x = \arctan \left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}} \right) \quad (5)$$

where a_x , a_y , and a_z are the accelerometer readings along the x, y, and z axes, respectively.

- Roll (θ_y): The rotation around the y-axis, calculated as:

$$\theta_y = \arctan \left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}} \right) \quad (6)$$

- Yaw (θ_z): The rotation around the z-axis, calculated using gyroscope data as:

$$\theta_z = \int \omega_z dt \quad (7)$$

where ω_z is the angular velocity around the z-axis.

The angular coordinates are combined to determine the orientation and position of the hand in 3D space. The transformation matrix for orientation is:

$$T = \begin{bmatrix} c_z c_y & c_z s_y s_x - s_z c_x & c_z s_y c_x + s_z s_x \\ s_z c_y & s_z s_y s_x + c_z c_x & s_z s_y c_x - c_z s_x \\ -s_y & c_y s_x & c_y c_x \end{bmatrix} \quad (8)$$

where

$$\begin{aligned} c_x &= \cos \theta_x; & s_x &= \sin \theta_x \\ c_y &= \cos \theta_y; & s_y &= \sin \theta_y \\ c_z &= \cos \theta_z; & s_z &= \sin \theta_z \end{aligned}$$

This matrix provides the hand's orientation in 3D space, which can then be used to identify specific gestures or positions.

IV. PROPOSED SYSTEM

The flowchart in Fig. 1 represents a basic workflow for a sign language recognition and conversion system. The user performs a gesture or hand movement that corresponds to a sign in the sign language. The movement is detected by sensors (e.g., flex sensors, accelerometers, or cameras). The sensor data is processed to determine whether it matches a predefined sign in the system's database. If the gesture does not match any known sign (False path), the system loops back to capture new input. If the gesture is recognized (True path), the process continues to the next step. The recognized sign

is converted into corresponding text or speech using text-to-speech algorithm [19], [20]. The process ends after the recognized sign is converted and output.

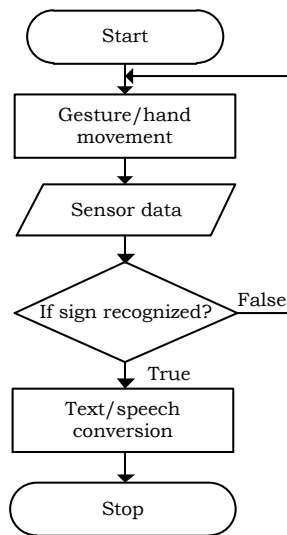


Fig. 1: Flowchart of the proposed work.

The Table II presents the accuracy of gesture recognition for different hand gestures under various conditions, highlighting the performance of different sensor types and environments. The alphabet “A” in American sign language has a gesture that corresponds to bending of 4 fingers excluding the thumb (Fig. 2). The resistance of the 4 sensors that is bent is recorded and the data is displayed on the LCD. This gesture achieved the highest accuracy, suggesting that flex sensors are highly effective for recognizing simple and distinct gestures, such as bending a single finger. The controlled environment likely minimizes external factors that could interfere with sensor performance, contributing to this high accuracy. The alphabet “B” in American sign language has a gesture that corresponds to bending of the thumb finger only (Fig. 3). The resistance of this sensor is recorded, and the data is displayed on the LCD. Gesture B also uses flex sensors and performs well with an accuracy of 92.5%. The slight decrease in accuracy might be due to the subtler motion of the thumb compared to a more pronounced single finger bend.

Table III presents the voltage value combinations used to represent the alphabets A, B, C, D, E, F, and G. Similarly, distinct sets of voltage values are employed to identify the other alphabets. On bending the first finger (Fig. 4), the resistance of the same changes and due to this variation pre-defined data is displayed on the LCD. This gesture involves multiple fingers bending, which could introduce more complexity in terms of sensor readings. The addition of accelerometers, alongside flex sensors, suggests that capturing more data from different angles improves recognition. On bending the first finger and second finger (Fig. 5), the resistance of the same changes

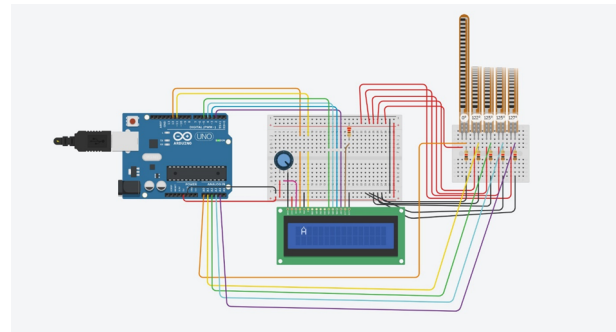


Fig. 2: Sign language for ‘A’ using proposed work.

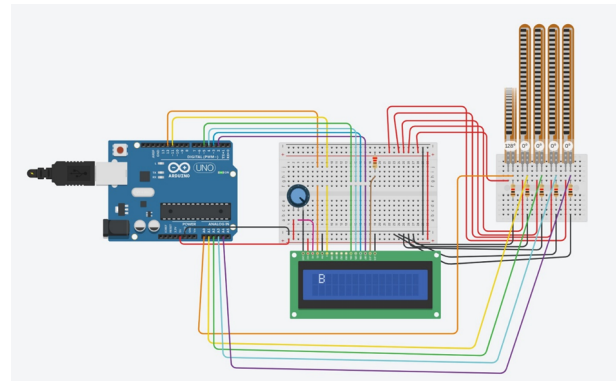


Fig. 3: Sign language for ‘B’ using proposed work.

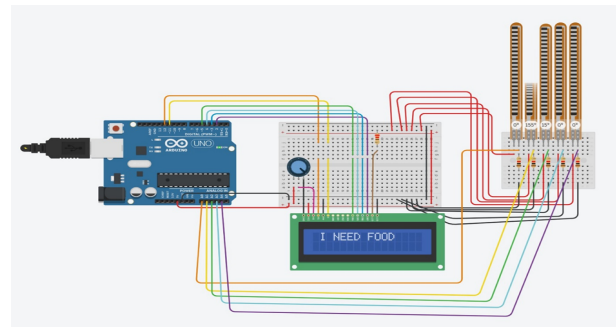


Fig. 4: Sign language for ‘I Need Food’ using proposed work.

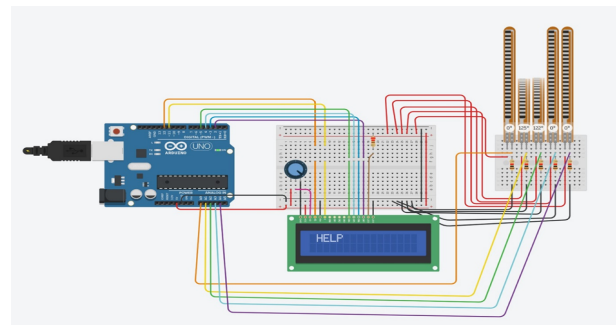


Fig. 5: Sign language for ‘Help’ using proposed work.

and due to this variation pre-defined data is displayed on the LCD. This gesture is more complex, as it involves an

TABLE II: Accuracy of gesture recognition under various conditions.

Gesture	Accuracy (%)	Sensor Type	Environment
A (Single Finger Bend)	95.2	Flex Sensors	Controlled Environment
B (Thumb Bend)	92.5	Flex Sensors	Controlled Environment
I Need Food (Multiple Finger Bend)	88.7	Flex Sensors, Accelerometers	Controlled Environment
Help (Open Hand Gesture)	90.3	Flex Sensors, Accelerometers	Real-World Setting

TABLE III: Flex sensor voltage values for alphabet recognition.

Index Reading	Middle Reading	Ring Reading	Pinky Reading	Bend Angle (°)	Detected Character
> 110	> 5	5 – 90	> 140	85	A
250	250	> 190	2	5	B
< 140	< 140	< 60	< 90	35	C
2	> 90	> 140	3	65	D
> 190	< 190	60 – 140	> 110	50	E
140 – 190	> 190	< 90	< 60	20	F
< 60	60 – 110	110 – 190	> 140	70	G

open hand, which requires precise tracking of finger positions and overall hand shape. Despite using both flex sensors and accelerometers, the accuracy drops slightly to 90.3%, possibly due to environmental factors in a real-world setting, such as ambient lighting or noise that may interfere with sensor data.

V. CONCLUSIONS

Deaf-mute individuals often face challenges in communicating with those who are unfamiliar with sign language. To address this issue, this work introduces a sensor-based glove system designed specifically for deaf-mute individuals. By relying on a minimal set of components, the system offers a more cost-effective alternative to vision-based gesture recognition methods. The gloves, embedded with resistors and sensors, enable users to perform hand gestures. The system first translates these gestures into corresponding text and then converts the text into speech using a text-to-speech synthesizer. The proposed approach can be extended to recognize words, phrases, and simple sentences by combining individual alphabet gestures. The use of additional short flex sensors can also expand its capability to identify a wider range of gestures.

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